

A recent development in a random walk with dependent steps and its limiting process

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Long-range dependence

- Long-range dependence (LRD): strong correlation in a sequence of random variables such that its covariance function is not summable or decays with a power law; it was first observed by Hurst (1951).

$$C(t) \sim ct^{-\alpha}$$

as $t \rightarrow \infty$, for $\alpha \in (0, 1)$ and a constant c .

Generalized Bernoulli process (GBP)

A generalized Bernoulli process

The GBP is defined in Lee (2021) as follows.

$$P(\xi_i = 1) = p, P(\xi_i = 0) = 1 - p,$$
$$P(\xi_{i_0} = 1, \xi_{i_1} = 1) = p(p + c|i_1 - i_0|^{2H-2}),$$

and in general,

$$P(\xi_{i_0} = 1, \xi_{i_1} = 1, \dots, \xi_{i_n} = 1) = p \prod_{j=1}^n (p + c|i_j - i_{j-1}|^{2H-2}),$$

for $0 < i_0 < i_1 < \dots < i_n$.

A generalized Bernoulli process

By the inclusion-exclusion principle, for any disjoint sets $A, B \subset \mathbb{N}$,

$$P(\cap_{i' \in B} \{\xi_{i'} = 0\} \cap_{i \in A} \{\xi_i = 1\}) = \sum_{k=0}^{|B|} \sum_{\substack{B' \subset B \\ |B'|=k}} (-1)^k P(\cap_{i \in B' \cup A} \{\xi_i = 1\}),$$

$$P(\cap_{i' \in B} \{\xi_{i'} = 0\}) = 1 + \sum_{k=1}^{|B|} \sum_{\substack{B' \subset B \\ |B'|=k}} (-1)^k P(\cap_{i \in B'} \{\xi_i = 1\}).$$

$\Rightarrow \{\xi_i, i \in \mathbb{N}\}$ is a stationary binary sequence with $P(\xi_i = 1) = p$, $P(\xi_i = 0) = 1 - p$, and covariance

$$\text{Cov}(\xi_i, \xi_j) = pc|i - j|^{2H-2}, \quad i \neq j.$$

A different version of GBP, GBP-II, is defined in Lee (2025). Let

$$p_n = \lambda n^{2H-2},$$

$$P(\xi_{i_0}^{(n)} = 1) = p_n,$$

$$P(\xi_{i_0}^{(n)} = 1, \xi_{i_1}^{(n)} = 1) = p_n c |i_1 - i_0|^{2H-2} \text{ for } 1 \leq i_0 < i_1 \leq n, \text{ and}$$

$$P(\xi_{i_0}^{(n)} = 1, \xi_{i_1}^{(n)} = 1 \cdots \xi_{i_k}^{(n)} = 1) = p_n c^k |(i_1 - i_0)(i_2 - i_1) \cdots (i_k - i_{k-1})|^{2H-2},$$

for any $1 \leq i_0 < i_1 < i_2 < \cdots < i_k \leq n$.

$\Rightarrow \{\xi_i^{(n)}, i = 1, \dots, n\}$ is a stationary process with $P(\xi_i^{(n)} = 1) = p_n$,
 $P(\xi_i^{(n)} = 0) = 1 - p_n$, and

$$\lim_{n \rightarrow \infty} \text{Corr}(\xi_i^{(n)}, \xi_j^{(n)}) = c |i - j|^{2H-2}, \quad i \neq j.$$

Fractional binomial distributions

Theorem (Lee (2025))

i) For the fractional binomial I, $E(B_n) = np$, and as $n \rightarrow \infty$, $E(B_n^k) \sim n^k p^k$, for $k \geq 2$, and the central moments are

$$E((B_n - np)^2) \sim \begin{cases} b_2^{(1)} n & \text{if } H \in (0, .5), \\ b_2^{(2)} n \ln n & \text{if } H = .5, \\ b_2^{(3)} n^{2H} & \text{if } H \in (.5, 1), \end{cases}$$
$$E((B_n - np)^k) \sim b_k n^{2H-2+k} \text{ for } H \in (0, 1),$$

for $k = 3, 4, \dots$.

ii) For the fractional binomial II, as $n \rightarrow \infty$,

$$E((B_n^\circ)^k) \sim c_k n^{(2H-1)k},$$

for $k \in \mathbb{N}$.

GBP-II VS fractional Poisson process

Theorem (Lee (2025))

For the fractional binomial distribution II^ with parameters H, c , and the fractional Poisson process with parameters $\mu = 2H - 1, \nu = c\Gamma(2H - 1)$,*

$$B_n^{\circ,*} / n^\mu \xrightarrow{d} X_\mu,$$

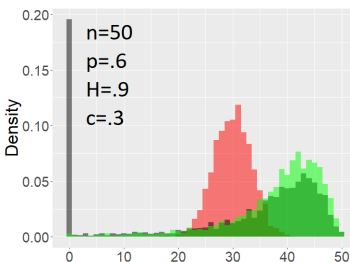
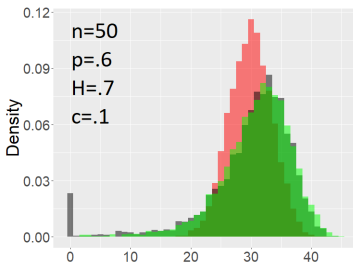
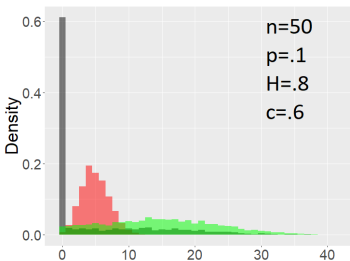
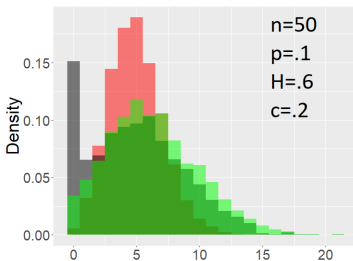
$$N_{\mu,\nu}(n) / n^\mu \xrightarrow{d} X_\mu,$$

as $n \rightarrow \infty$, where X_μ is the second family of Mittag-Leffler distribution whose mgf is Mittag-Leffler function, i.e.,

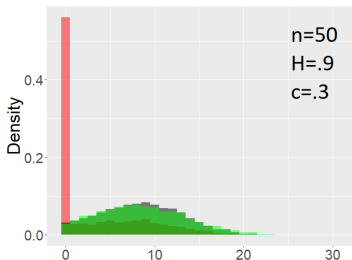
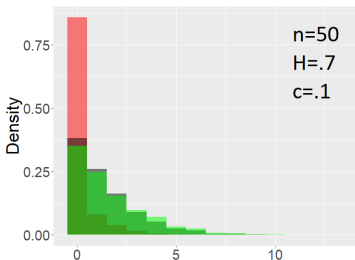
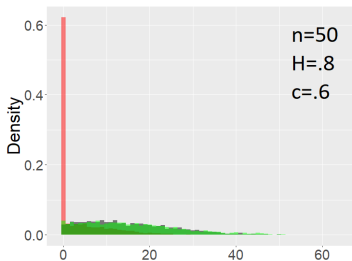
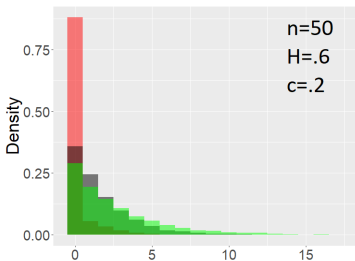
$$E(e^{tX_\mu}) = E_\mu(\nu t) \text{ for } t \in \mathbb{R},$$

where

$$E_\mu(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\mu k + 1)}.$$



■ Binomial
 ■ Fractional binomial I
 ■ Fractional binomial I*



■ Fractional binomial II
 ■ Fractional binomial II*
 ■ Fractional Poisson

For any $T \geq 1, n \in \mathbb{N}$, define a sequence of dependent Bernoulli random variables $\{\xi_i^{(n)}; i \in [\lfloor nT \rfloor]\}$ with a function f_n as follows (Lee, 2026).

$$P(\xi_i^{(n)} = 1) = bf_n(\lfloor nT \rfloor),$$

and

$$P(\cap_{j \in [k]} (\xi_{ij}^{(n)} = 1)) = bf_n(\lfloor nT \rfloor) \prod_{j \in [k-1]} f_n(|i_{j+1} - i_j|), \quad (1)$$

for $1 \leq i_1 < i_2 < \dots < i_k \leq \lfloor nT \rfloor$ and $b \in (0, 1]$. For any disjoint sets $A_0, A_1 \subset [\lfloor nT \rfloor]$,

$$\begin{aligned} & P(\cap_{i' \in A_0} \{\xi_{i'}^{(n)} = 0\} \cap_{i \in A_1} \{\xi_i^{(n)} = 1\}) \\ &= \sum_{k=0}^{|A_0|} \sum_{\substack{F' \subset A_0 \\ |F'|=k}} (-1)^k P(\cap_{i \in F' \cup A_1} \{\xi_i^{(n)} = 1\}). \end{aligned} \quad (2)$$

Assumption (A)

- i) $f_n : \mathbb{N} \rightarrow (0, 1)$ is a decreasing function such that $f_n(x+1)/f_n(x)$ is non-decreasing as x increases, and
- ii) $f_n(2) > f_n(1)^2$.

Example

- i) $f_n(x) = c \exp(-\lambda x/n)$, for $c \in (0, 1)$, $\lambda > 0$.
- ii) $f_n(x) = cx^{\alpha-1}$ for $c \in (0, 2^{\alpha-1})$, $\alpha < 1$.
- iii)

$$f_n(x) = c \exp(-\lambda x/n) x^{\alpha-1}$$

where $c \in (0, 2^{\alpha-1})$, $\alpha \leq 1$, $\lambda > (\alpha - 1)/T$.

Scaling limit of dependent random walk

Let $\xi_i^{(n)} \sim \text{GBP}(f_n)$, where

$$f_n(x) = c \exp(-\lambda x/n) x^{\alpha-1},$$

and $c \in (0, 2^{\alpha-1})$, $\alpha \in (0, 1]$, $\lambda > (\alpha - 1)/T$.

$$X_t^{(n)} = \frac{\sum_{i=1}^{\lfloor nt \rfloor} \xi_i^{(n)}}{n^\alpha} \xrightarrow{d} X_t$$

and

$$X_t^{(*,n)} = \frac{\sum_{i=1}^{\lfloor nt \rfloor} \xi_i^{(*,n)}}{n^\alpha} \xrightarrow{d} X_t^*$$

in the space $D[0, T]$ and $D[0, \infty)$, respectively, with the Skorokhod (J_1) topology. We call X_t and X_t^* **exponential Mittag-Leffler (EML) processes**.

$$X_t \sim \text{EML}(\alpha, \lambda, b, T) \quad X_t^* \sim \text{EML}^*(\alpha, \lambda)$$

Self-similar property of EML processes:

For any $u > 0$,

$$\{X_{\textcolor{red}{u}t}(\alpha, \lambda, b, T); t \in [0, T/u]\} \stackrel{f.d.d}{=} \{\textcolor{red}{u}^\alpha X_t(\alpha, \textcolor{red}{u}\lambda, b, T/\textcolor{red}{u}); t \in [0, T/u]\},$$

and

$$\{X_{\textcolor{red}{u}t}^*(\alpha, \lambda); t > 0\} \stackrel{f.d.d}{=} \{\textcolor{red}{u}^\alpha X_t^*(\alpha, \textcolor{red}{u}\lambda); t > 0\}.$$

The pdfs of X_t and X_t^* solve **tempered fractional** diffusion equations,

$${}_0D_t^{\alpha,\lambda} p^*(x, t) = -\frac{\partial p^*(x, t)}{\partial x} + \delta(x) {}_0D_t^{\alpha,\lambda}(1),$$

$${}_0D_t^{\alpha,\lambda} p(x, t) = -\frac{\partial p(x, t)}{\partial x} + \left(\delta(x) {}_0D_t^{\alpha,\lambda}(1) + \delta'(x) - \delta'(x) a_T {}_0D_t^{\alpha,\lambda}(t) \right),$$

where ${}_0D_t^{\alpha,\lambda}$ is the Riemann-Liouville tempered fractional derivative

$${}_0D_t^{\alpha,\lambda} f(t) = \frac{e^{-\lambda t}}{\Gamma(n-\alpha)} \frac{d^n}{dt^n} \int_0^t (t-s)^{-\alpha+n-1} e^{\lambda s} f(s) ds, \quad n-1 \leq \alpha < n.$$

- The pdf of X_t^* is

$$p^*(x, t) = e^{-\lambda t} h(x, t) + \lambda \int_0^t e^{-\lambda y} h(x, y) dy$$

for $t, x \geq 0$, where h is the pdf of inverse stable subordinator.

- The pdf of X_t is

$$p(x, t) = -a_T \int_0^t \frac{\partial p^*(x, y)}{\partial x} dy + \delta(x).$$

X_t has stationary increments, $\nabla X_t = X_t - X_{t-d}$, and

$$E(\nabla X_s \nabla X_{s+t}) \sim c' \Delta^2 e^{-\lambda t} t^{\alpha-1}$$

as $d = \Delta \rightarrow 0$. For any $m \in \mathbb{N}$, $0 < t_1 < t_2 < \cdots < t_m \leq T$, and $k_i \in \mathbb{N}, i = 1, 2, \dots, m$,

$$\lim_{d \rightarrow 0} \frac{E(\prod_{i=1}^m (\nabla X_{t_i})^{k_i})}{\prod_{i=1}^m E((\nabla X_{t_i})^{k_i})} = (c/a_T)^{m-1} e^{-\lambda(t_m - t_1)} \prod_{i=2}^m (t_i - t_{i-1})^{\alpha-1}.$$

Let $\nabla X_t^* = X_t^* - X_{t-d}^*$ be the increment of X_t^* .

$$\mathbb{E}(\nabla X_s^* \nabla X_{s+t}^*) \sim c' \Delta^2 e^{-\lambda(s+t)} s^{\alpha-1} t^{\alpha-1}$$

as $d = \Delta \rightarrow 0$. For any $m \in \mathbb{N}$, $0 = t_0 < t_1 < t_2 < \dots < t_m$, and $k_i \in \mathbb{N}$, $i = 1, 2, \dots, m$,

$$\mathbb{E}\left(\prod_{i=1}^m (\nabla X_{t_i}^*)^{k_i}\right) \sim g(k_1, \dots, k_m) \Delta^{\sum_{i \in [m]} (\alpha(k_i-1)+1)} e^{-\lambda t_m} \prod_{i=1}^m (t_j - t_{j-1})^{\alpha-1}.$$

Exponential process

Let $\xi_i^{(n)} \sim \text{GBP}(f_n)$ where $f_n(x) = c \exp(-\lambda x/n)$, $c \in (0, 1)$, $\lambda > 0$,

- the pdf of X_t^* is

$$p^*(x, t) = I_{(0,t]}(x) \lambda e^{-\lambda x} + \delta_t(x) e^{-\lambda t},$$

- the pdf of X_t is

$$p(x, t) = b e^{-\lambda T} \left(I_{(0,t]}(x) \lambda (2 + \lambda t - \lambda x) e^{-\lambda x} + \delta_t(x) e^{-\lambda t} \right) \\ + (1 - b e^{-\lambda T} - b e^{-\lambda T} t \lambda) \delta(x).$$

- X_t has stationary increments such that

$$E(\nabla X_s \nabla X_{s+t}) \sim c' \Delta^2 e^{-\lambda t},$$

as $d = \Delta \rightarrow 0$.

Mittag-Leffler process

If $f_n(x) = cx^{\alpha-1}$, where $c \in (0, 1), \alpha \in (0, 1)$

- the pdfs of X_t^* and X_t solve **time-fractional** diffusion equations,

$$\partial_t^\alpha p^*(x, t) = -\frac{\partial}{\partial x} p^*(x, t),$$

$$\partial_t^\alpha p(x, t) = -\frac{\partial}{\partial x} p(x, t) + \delta'(x) \left(1 - \frac{c' t^{1-\alpha}}{\Gamma(2-\alpha)} \right),$$

where ∂_t^α is the Caputo fractional derivative.

- X_t has stationary increments with

$$E(\nabla X_s \nabla X_{s+t}) \sim c' \Delta^2 t^{\alpha-1}.$$

$$E\left(\prod_{i=1}^m (\nabla X_{t_i})^{k_i}\right) \sim \prod_{i=1}^m E((\nabla X_{t_i})^{k_i}) \prod_{i=2}^m \frac{(t_i - t_{i-1})^{\alpha-1}}{c'}.$$

THANK YOU!